Advanced Econometrics for Data Science – Lecture 08

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Example 1

```
# ---------------------------------------------------------------------
# Example 1
# Simple Monte Carlo experiment
# What estimate should we get?
# ---------------------------------------------------------------------

# specification
N.iter = 1000
N = 1000
beta0 = 1
beta1 = 2
beta1.hat.vec = rep(NA, times=N.iter)

# Monte Carlo Loop
for(iter in 1:N.iter) {
  x = rnorm(n=N, mean=10, sd=1)
  y = beta0 + beta1*x + rnorm(n=N, mean=0, sd=1)
  model.iter = lm(y~x)
  beta1.hat.vec[iter] = model.iter$coefficients[2]
  rm(x,y)
}

hist(beta1.hat.vec)
```

```
# -------------------------------------------------------------------
# Example 2
# What pvalues should we get?
# -------------------------------------------------------------------

library("car")
# specification
N.iter = 1000
N = 1000
beta0 = 1
beta1 = 2
beta1.pvalues = rep(NA, times=N.iter)

# Monte Carlo Loop
for(iter in 1:N.iter) {
  x = rnorm(n=N, mean=10, sd=1)
  y = beta0 + beta1*x + rnorm(n=N, mean=0, sd=1)
  model.iter = lm(y~x)
  verif = linearHypothesis(model.iter, c("x=2"))
  beta1.pvalues[iter] = verif$`Pr(>F)`[2]
  rm(x,y)
}

hist(beta1.pvalues)
# Does it look as it should?
# How many rejections do we have?
mean(beta1.pvalues<0.05)
```

```
# -------------------------------------------------------------------
# Example 3
# A model with two independent variables
# H0: x=2 and z=3
# -------------------------------------------------------------------
library("car")
# specification
N.iter = 1000
N = 1000
beta0 = 1
beta1 = 2
beta2 = 3
H0.pvalues = rep(NA, times=N.iter)

# Monte Carlo Loop
for(iter in 1:N.iter) {
  x = rnorm(n=N, mean=10, sd=1)
  z = rpois(n=N, lambda=5)
  y = beta0 + beta1*x + beta2*z + rnorm(n=N, mean=0, sd=1)
  model.iter = lm(y~x+z)
  verif = linearHypothesis(model.iter, c("x=2", "z=3"))
  H0.pvalues[iter] = verif$`Pr(>F)`[2]
  rm(x,y)
}
hist(H0.pvalues)
# How many rejections do we have?
mean(H0.pvalues<0.05)
```

```
# -------------------------------------------------------------------
# Example 4
# A model with two independent variables
# H0: x=2 and z=3
# but now we will use simple hypothesis testing
# -------------------------------------------------------------------

library("car")

# specification
N.iter = 1000
N = 1000
beta0 = 1
beta1 = 2
beta2 = 3
if.rejected = rep(NA, times=N.iter)

# Monte Carlo Loop
for(iter in 1:N.iter) {
  x = rnorm(n=N, mean=10, sd=1)
  z = rpois(n=N, lambda=5)
  y = beta0 + beta1*x + beta2*z + rnorm(n=N, mean=0, sd=1)
  model.iter = lm(y~x+z)
  verif.x = linearHypothesis(model.iter, c("x=2"))
  verif.z = linearHypothesis(model.iter, c("z=3"))
  if.rejected[iter] = (verif.x$`Pr(>F)`[2]<0.05 | verif.z$`Pr(>F)`[2]<0.05)
  rm(x,y,z)
}

table(if.rejected)
# How many rejections do we have?
mean(if.rejected==1)
```

Job prestige regression on zodiac signs

	Coef.	Std. Err.	t	Pr(|t|>t*)
zodiac2	-1.186	1.189	-1.00	0.319
zodiac3	-1.163	1.215	-0.96	0.338
zodiac4	-1.944	1.225	-1.59	0.113
zodiac5	-0.844	1.232	-0.69	0.493
zodiac6	0.490	1.249	0.39	0.695
zodiac7	-1.736	1.233	-1.41	0.159
zodiac8	-2.517	1.255	-2.01	0.045
zodiac9	-1.772	1.303	-1.36	0.174
zodiac10	-1.710	1.208	-1.42	0.157
zodiac11	-0.063	1.216	-0.05	0.959
zodiac12	-0.446	1.186	-0.38	0.707
constant	39.468	0.866	45.60	0.000

Source: [2] p. 190

How to select useful empirical model?

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- There are four potential basic mistakes in selecting a model from data evidence:
  1. mis-specifying the general unrestricted model;
  2. failing to retain variables that should be included;
  3. retining variables that should be omitted; and
  4. selecting a noncronguent representation, which renders conventional inference hazardous.

  From: J. Campos, N.R. Ericsson, D.F. Hendry,*General-to-specific Modeling: An Overview and Selected Bibliography*, Board of Governors of the Federal Reserve System, International Finance Discussion Papers, Number 838, August 2005.

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- General-to-specific approach – a possibility

General-to-specific modelling - [1] p. 3

- A viable methodology for empirical modeling is an integral component of achieving [a "good" empirical model of a DGP]. Despite the controversy surrounding every aspect of econometric methodology, the "LS" (or London School of Economics) approach has emerged as a leading methodology for empirical modeling.
- One of the LSE approach's main tenets is general-to-specific modeling, sometimes abbreviated as Gets.
- In general-to-specific modeling, empirical analysis starts with a general statistical model that captures the essential characteristics of the underlying dataset, i.e., that general model is congruent.
- Then, that general model is reduced in complexity by eliminating statistically insignificant variables, checking the validity of the reductions at every stage to ensure congruence of the finally selected model.

General-to-specific modelling - [1] p. 3

1. Ascertain that the general statistical model is congruent.
2. Eliminate a variable (or variables) that satisfies the selection (i.e., simplification) criteria.
3. Check that the simplified model remains congruent.
4. Continue steps 2 and 3 until none of the remaining variables can be eliminated.

Gets example

```
Fixed-effects (within) regression               Number of obs      =       630
------------------------------------------------------------------------------
     lcrmrte |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     lprbarr |  -.3917945   .0334336   -11.72   0.000     -.457473   -.3261161
    lprbconv |  -.3111651   .0218356   -14.25   0.000    -.3540598   -.2682704
    lprbpris |  -.2134135    .033476    -6.38   0.000    -.2791752   -.1476518
     lavgsen |   .0327006   .0259671     1.26   0.208    -.0183103    .0837115
      lpolpc |   .4253936   .0276187    15.40   0.000     .3711382    .4796489
    ldensity |  -.0437532   .2643179    -0.17   0.869    -.5629903     .475484
      ltaxpc |  -.0214103     .03942    -0.54   0.587    -.0988486     .056028
       lwcon |  -.0767867   .0381908    -2.01   0.045    -.1518102   -.0017632
       lwfir |  -.0345552   .0289553    -1.19   0.233    -.0914362    .0223258
       _cons |  -1.137501   .3002654    -3.79   0.000    -1.727355   -.5476472
------------------------------------------------------------------------------
```

Modelling - to sum up

Prescription for successful empirical econometric research:

1. think brilliantly
2. be infinitely creative
3. be outstandingly lucky
4. stick to being a theorist

[1] p. 34 or [3] p. 29-30

General advice on writing and publishing

http://www.principlesofeconometrics.com/poe5/writing.html

- Principles of Econometrics, Fifth Edition, Chapter 1.7
- Hal Varian (1997) ”How to build and economic model in your spare time”
- John Duffy "How to research and write a economics term paper"
- David Romer’s Rules
- Michael Kremer’s writing paper checklist
- Peter Kennedy’s Ten Commandments
- Pozostałe

David Romer's Rules

- Don't clutter up your life with other activities; **just write.**
- Don't carry out a thorough and comprehensive search of the literature; **just write.**
- Don't attempt to make sure that every page you write shows the full extent of your professional skills; **just write.**
- Don't write a well-organized, well-integrated, unified dissertation; **just write.**
- Don't think profound thoughts that shake the intellectual foundations of the discipline; **just write.**

Writing Papers: A Checklist – Michael Kremer

1. Describe the data set. Where is it from? How many observations?

2. Include a table showing the means of the variables, and the standard deviations.

3. As a rule, it is usually better to show standard errors, rather than t-statistics.

4. Look at a journal to see how tables are laid out. Avoid horizontal lines in tables.

5. Include basic information like sample size and R* in all tables.

6. To the extent possible, make tables self contained with clearly-explained titles and extensive notes.

7. Explain clearly where the identification comes from. As a disciplinary device, try t include an explanation that would be intelligible to a non-economist.

8. Did you drop any data? What rules did you use in deciding what data to drop?

My suggestion is rather different: I think that you should look for your ideas outside the academic journals—in newspapers, in magazines, in conversations, and in TV and radio programs. When you read the newspaper, look for the articles about economics . . . and then look at the ones that aren't about economics, because lots of the time they end up being about economics too. Magazines are usually better than newspapers because they go into issues in more depth. On the other hand, a shallower analysis may be more stimulating: there's nothing like a fallacious argument to stimulate research.[1]

Before you start trying to decide whether your idea is correct, you should stop to ask whether it is interesting. If it isn't interesting, no one will care whether it is correct or not. So try it out on a few people—see if they think that it is worth pursuing. What would follow from this idea if it is correct? Would it have lots of implications or would it just be a dead end?

1.3 Don't look at the literature too soon

The first thing that most graduate students do is they rush to the literature to see if someone else had this idea already. However, my advice is to wait a bit before you look at the literature. Eventually you should do a thorough literature review, of course, but I think that you will do much better if you work on your idea for a few weeks before doing a systematic literature search. There are several reasons for delay.

First, you need the practice of developing a model. Even if you end up reproducing exactly something that is in the literature already you will have learned a lot by doing it—and you can feel awfully good about yourself for developing a publishable idea! (Even if you didn't get to publish it yourself ...)

Second, you might come up with a different approach than is found in the literature. If you look at what someone else did your thoughts will be shaped too much by their views—you are much more likely to be original if you plunge right in and try to develop your own insights.

How to Research and Write an Economics Term Paper

No one can tell you how to research and write a term paper. Good research and writing skills require constant practice. Nonetheless, I would like to offer you some suggestions as to how you might approach your term paper assignment. While these are only suggestions, I encourage you to follow as many of these as possible.

1. Get started early. Do not wait until the last minute to complete this paper assignment. You (and I) will be sorry. The best way to avoid procrastinating is to take small steps toward completion of the paper. Go to the library or surf the Web looking for articles, books, or data on a topic that may interest you. Once you have taken a first small step you have, in effect, overcome your urge to procrastinate! After taking this first step, set some deadlines for the completion of increasingly larger tasks: paper topic, outline, research, data gathering and analysis, first draft, second draft etc.

8. What exactly are you trying to say?

If you are not sure, imagine the confusion that your reader faces. Before you start writing, give some thought to what it is you want to say. Then, while writing, make sure that every paragraph has a point. Finally, read what you have written out loud. Good writing has a conversational flow. This ow can often be achieved only by reading what you have written aloud and making sure that what you hear "sounds right."

9. Be objective.

President Harry S. Truman once complained that what he really needed was a "one-armed" economist. The reason? Economists are especially fond of saying "...on the other hand..." That is, economists like to consider all possibilities, especially those that other (rival) economists have overlooked.

Since the reader of your paper is an economist (me), you might want to bear in mind the economists' preference for objectivity. On the other hand (oops), you should avoid being too objective when your research or the data call for taking a strong stand on a particular issue. Another popular expression among economists is "it depends." Feel free to use this expression a lot. You will find that it frequently characterizes all kinds of economic phenomena.

SINNING IN THE BASEMENT: WHAT ARE THE RULES? THE TEN COMMANDMENTS OF APPLIED ECONOMETRICS

Peter E. Kennedy

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Abstract. Unpleasant realities of real-world data force applied econometricians to violate the prescriptions of econometric theory as taught by our textbooks. Leamer (1978) vividly describes this behavior as wanton sinning in the basement, with sinners' metamorphizing into high priests as they ascend to the third floor to teach econometric theory. But this sinning is not completely wanton — applied econometricians do (or should) follow some unwritten rules of behavior, in effect bounding the sinning and promoting a brand of honor among sinners. This paper exposits these rules, and culls from them an unauthorized list of the Ten Commandments of applied econometrics.

Keywords. Applied econometrics; Methodology; Data mining.

The Rules of Sinning

- Rule $\#1$: Use common sense and economic theory.
- Rule $\#2$: Avoid type III errors.
- Rule $\#4$: Inspect the data.
- Rule $\#5$: Keep it sensibly simple.
  This KISS rule, based on Zellner's (1992, 2001) 'keep it sophisticatedly simple' rule, should not, as Zellner warns, be confused with the commercial 'keep it simple, stupid' rule, because some simple models are stupid, containing logical errors or being at variance with facts.

The Rules of Sinning

- Rule #7: Understand the costs and benefits of data mining.
  This objection to data mining is the basis for complaints about Hendry's (1980, p. 403) edict that 'The three golden rules of econometrics are test, test, and test'. Applied indiscriminately, these rules, and their associated 'general to specific' specification search methodology, can lead to problems such as pretest bias and distortion of type I error rates, jokes about having more test statistics than observations, and reference to Ronald Coase's oft-cited comment that 'If you torture the data long enough, Nature will confess'.
- Rule #9: Do not confuse statistical significance with meaningful magnitude.
- Rule #10: Report a sensitivity analysis.

The Ten Commandments of Applied Econometrics

3. The Ten Commandments of Applied Econometrics

With apologies to Thomas (1976) and to Driscoll (1977), I have taken the liberty of culling from these rules Ten Commandments for applied econometricians.

1. **Thou shalt use common sense and economic theory.**
   Corollary: Thou shalt not do thy econometrics as thou sayest thy prayers.
2. **Thou shalt ask the right questions.**
   Corollary: Thou shalt place relevance before mathematical elegance.
3. **Thou shalt know the context.**
   Corollary: Thou shalt not perform ignorant statistical analyses.
4. **Thou shalt inspect the data.**
   Corollary: Thou shalt place data cleanliness ahead of econometric godliness.
5. **Thou shalt not worship complexity.**
   Corollary: Thou shalt not apply asymptotic approximations in vain.
   Corollary: Thou shalt not talk Greek without knowing the English translation.

The Ten Commandments of Applied Econometrics

6. **Thou shalt look long and hard at thy results.**
   Corollary: Thou shalt apply the laugh test.
7. **Thou shalt beware the costs of data mining.**
   Corollary: Thou shalt not worship R^2.
   Corollary: Thou shalt not hunt statistical significance with a shotgun.
   Corollary: Thou shalt not worship the 0.05% significance level.
8. **Thou shalt be willing to compromise.**
   Corollary: Thou shalt not worship textbook prescriptions.
9. **Thou shalt not confuse significance with substance.**
   Corollary: Thou shalt not ignore power.
   Corollary: Thou shalt not test sharp hypotheses.
   Corollary: Thou shalt seek additional evidence.
10. **Thou shalt confess in the presence of sensitivity.**
    Corollary: Thou shalt anticipate criticism.

Stephen King's 10% Rule.

From Stephen King's *On Writing*:

"In the spring of my senior year at Lisbon High—1966, this would have been—I got a scribbled comment that changed the way I rewrote my fiction once and forever. Jotted below the machine-generated signature of the editor was this mot: "Not bad, but PUFFY. You need to revise for length. Formula: 2nd Draft = 1st Draft – 10%. Good luck."

"I wish I could remember who wrote that note—Algis Budrys, perhaps. Whoever it was did me a hell of a favor. I copied the formula out on a piece of shirt-cardboard and taped it to the wall beside my typewriter. Good things started to happen for me shortly thereafter."

John Grisham's Do's And Don'ts For Popular Fiction.

John Grisham is another popular writer who takes the *less is more* approach.

"Read each sentence at least three times in search of words to cut."

"Most writers use too many words, and why not? We have unlimited space and few constraints."

George Orwell Concurs.

"If it is possible to cut a word out, always cut it out."

Keep the ass to the chair.
-James Buchanan

Zadanie 22

Na święta choinka została ubrana tylko w bombki. Dobrać najlpiej dopasowaną hiperpłaszczyznę do bombek pod względem kryterium błędu średniokwadratowego. Rozpatrzyć różne przypadki.

Bibliography

[1] J. Campos, N.R. Ericsson, D.F. Hendry,*General-to-specific Modeling: An Overview and Selected Bibliography*, Board of Governors of the Federal Reserve System, International Finance Discussion Papers, Number 838, August 2005.

[2] J. Mycielski, *Ekonometria*, textbook for WNE UW students [in Polish], University of Warsaw, 2009.

[3] Hendry D.F., *Econometric Methodology: A Personal Perspective*, Chpter 10 in T.F. Bewley (ed.) *Advances in Econometrics: Fifth World Congress*, Volume 2, Cambridge University Press, Cambridge, 29-48.